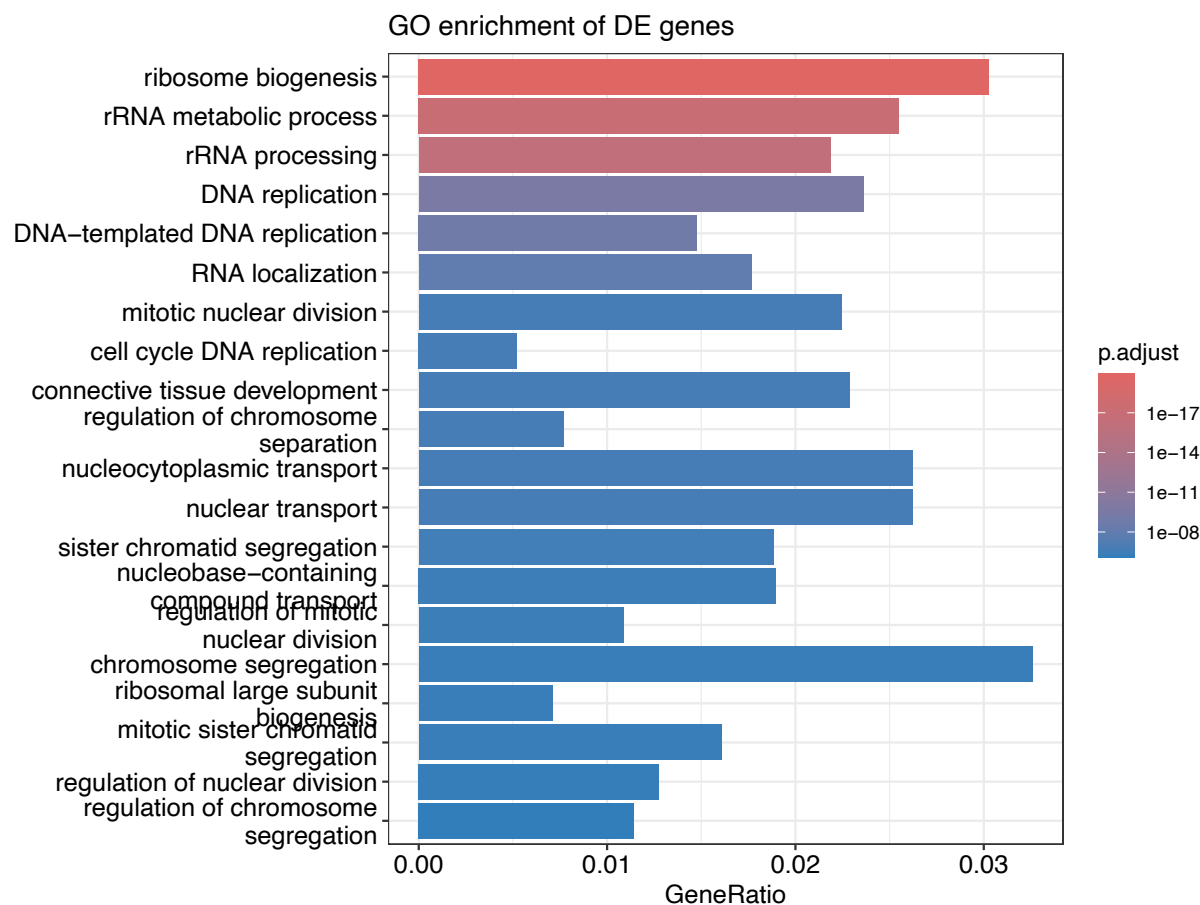
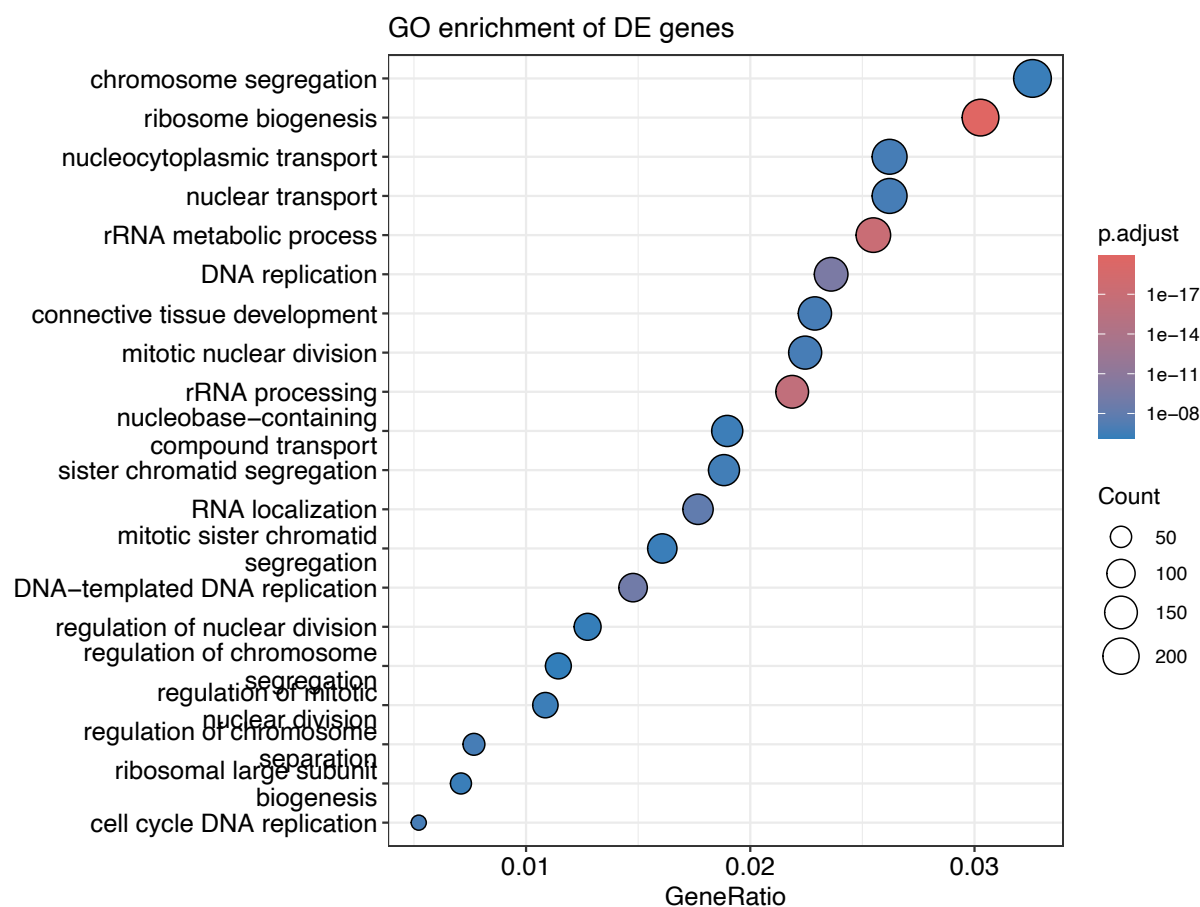
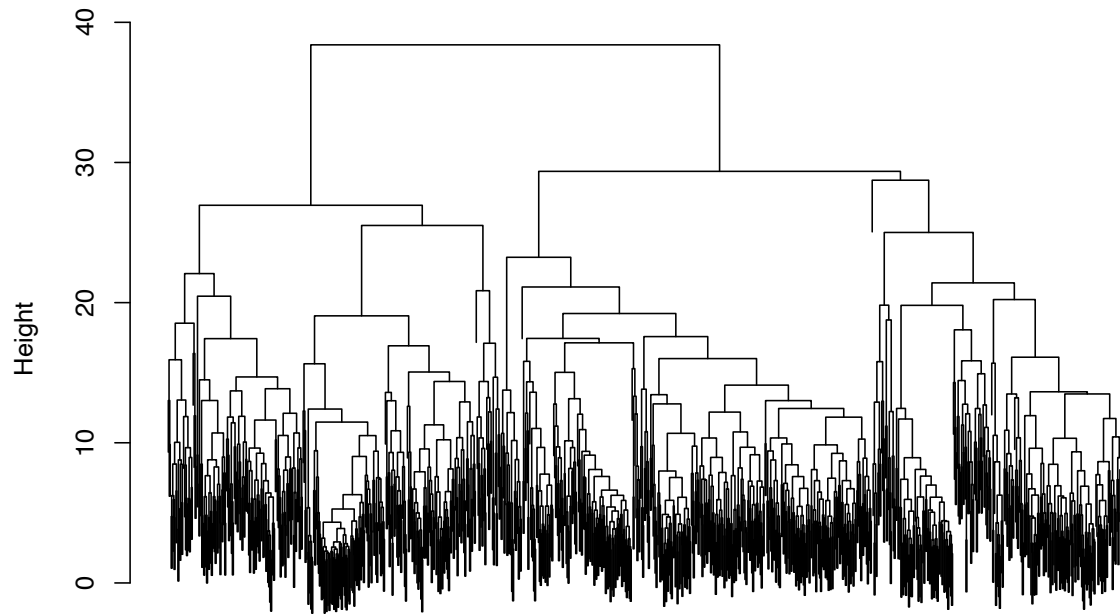


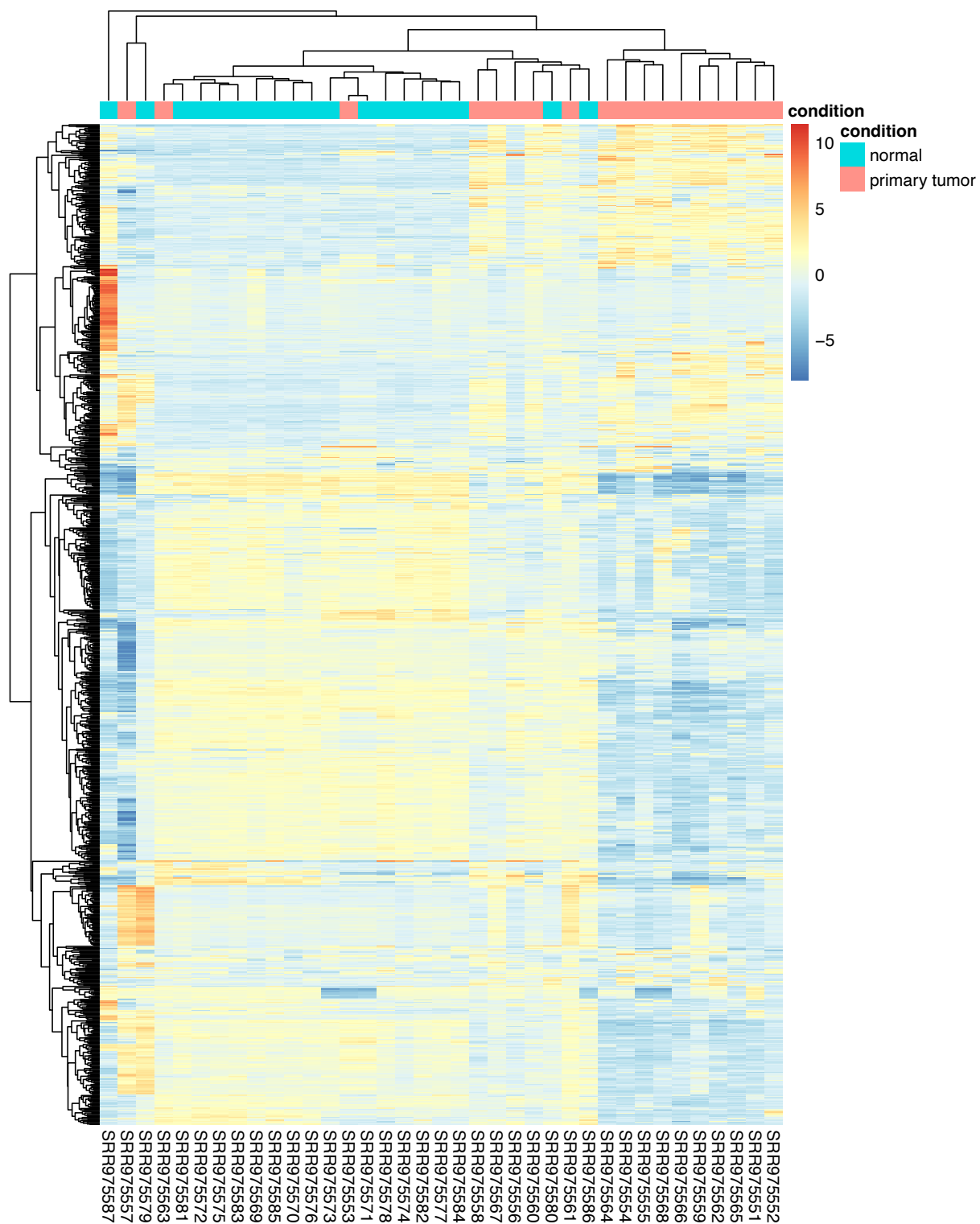
Lab 6

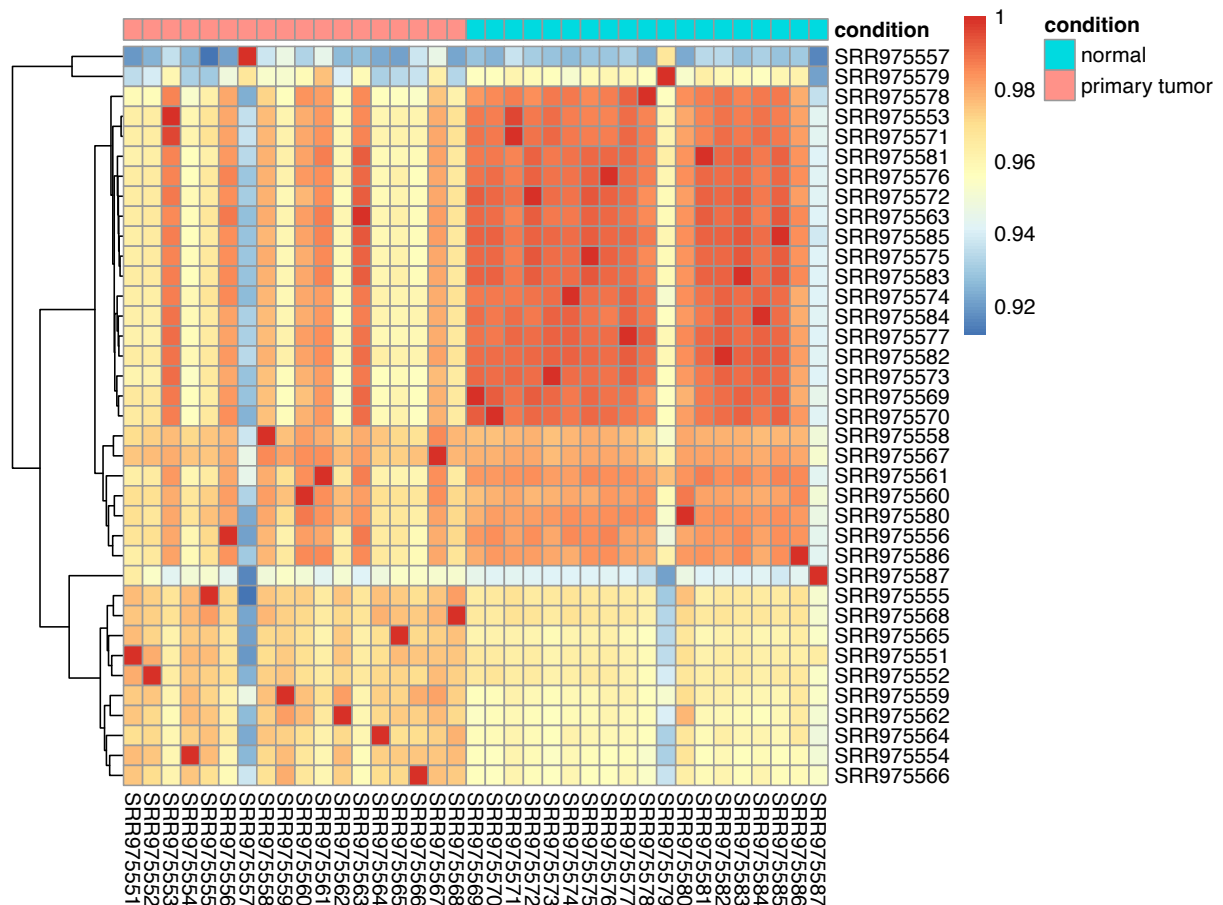


Hierarchical clustering of top variable genes



dist_genes
hclust (*, \"complete\")





3. Gene set enrichment analysis

```
1 head(GO_df$Description, 10)
```

```
1 [1] "ribosome biogenesis" "rRNA metabolic process"
2 [3] "rRNA processing" "DNA replication"
3 [5] "DNA-templated DNA replication" "RNA localization"
4 [7] "mitotic nuclear division" "cell cycle DNA replication"
5 [9] "connective tissue development" "regulation of chromosome separation"
```

The dominant functional themes that distinguish **primary tumor** from **normal** tissue are mainly related to **ribosome biogenesis, RNA metabolism, and cell cycle / DNA replication**.

These enriched functions indicate that tumor samples show increased activity in protein synthesis machinery (ribosome and rRNA-related processes) and strong up-regulation of cell cycle and DNA replication pathways.